

If we view the map $\mathcal{F}_t$ as a change of variables, we have

$$\text{Vol}(\mathcal{S}(t)) = \int_{\mathcal{S}} |D| d\omega,$$

where $D = \det \left(\frac{\partial \mathcal{F}_t}{\partial \mathbf{z}} \right)$.

To understand where Liouville's theorem comes from, recall that the variational equations of the last chapter are a system of ordinary differential equations for $\mathbf{W}(t) = \mathcal{F}'_t(\mathbf{z}(t))$:

$$\frac{d\mathbf{W}}{dt} = \mathbf{f}'(\mathbf{z}(t))\mathbf{W}.$$

Thus

$$\dot{\mathbf{W}}\mathbf{W}^{-1} = \mathbf{f}'(\mathbf{z}(t)).$$

Now let $D = \det(\mathbf{W})$. One can show (see Exercise 5) that

$$\frac{\dot{D}}{D} = \text{tr}(\dot{\mathbf{W}}\mathbf{W}^{-1}).$$

This implies that

$$\dot{D} = \text{div}(\mathbf{f}(\mathbf{z}(t)))D,$$

and thus

$$D(t) = D(0)e^{\int_0^t \text{div}(\mathbf{f}(\mathbf{z}(s)))ds}.$$

In particular, if $\text{div} \mathbf{f} \equiv 0$, we see that $D \equiv D(0) = 1$ and it follows that the volume is constant.

Liouville's theorem may be summarized compactly as:

$$\nabla \cdot \mathbf{f} = 0 \Rightarrow \det \mathcal{F}'_t = 1.$$

Example 2.3 A 1-d oscillator with Lennard-Jones potential is described by the equations

$$\begin{aligned} \dot{q} &= p, \\ \dot{p} &= -\phi'_{\text{LJ}}(q). \end{aligned}$$